a result, machines can identify the most meaningful combinations of words to form a sentence.

Transformers revolutionized natural language processing by enhancing both efficiency and accuracy in understanding textual data. They simplify key computation-intensive tasks by reducing the need for large amounts of data. By using an attention mechanism, they help put the meaning of a word in context, considering the words before and after it. Therefore, computers can generate long, semantically sound sentences using only small prompts.

Take, for instance, the task of completing the phrase "green eggs and . . .." Older natural language processing models often generate responses like "green eggs and unicorns" or "green eggs and spaceships." While grammatically correct, these completions failed to capture the contextual understanding required to produce meaningful and contextually relevant responses based on the original prompt.

The newer class of models has made significant strides in overcoming these limitations. Not only can they form grammatically valid sentences, but they can also provide more contextually relevant answers based on vast amounts of public literature. Today, when prompted with "green eggs and . . .," a model is likely to continue with "ham," referencing the well-known Dr. Seuss book. This demonstrates the model's ability to draw upon extensive knowledge and cultural references to generate responses that are not only syntactically correct but also semantically meaningful and contextually appropriate.

The cool thing about this technology is that it's not confined to text generation tasks; it also finds applications in computer vision for tasks like object detection and image classification, as well as in speech recognition for AI voice assistants. Any media for which feature extraction and semantic understanding are crucial can benefit from the capabilities of transformers.

1.6 Handling the concerns and implications of generative AI

As with any powerful tool, generative AI comes with significant responsibilities and valid concerns. To keep things simple, I will refrain from engaging in abstract philosophical debates about the ethical use of AI and instead provide practical guidelines for the responsible and effective utilization of generative AI in web applications.

1.6.1 What are the limitations of generative AI?

Some of the most crucial challenges and limitations to generative AI include

- *Quality control problems*—These concerns are about ensuring the quality and reliability of generated outputs, especially their accuracy, coherence, and relevance. There is always a chance that the output will be inadequate or unacceptable, which poses a risk of causing more harm than good.
- *Resource intensiveness*—Training our own models for generative AI demands computational resources and infrastructure for training and inference, which can be costly and resource intensive. Not everyone has the budget to run large, expensive GPUs for extended periods of time.